

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07
COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I K. Fogest (192525035) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

K-J08087
Signature of the student:

① DESIGN THE TRANSPORT

Given:

File = 150 MB

segment = 1500 bytes

overhead = 40 bytes

Bandwidth = 50 mbps.

Reliable transport layer
uses acknowledgement,
sequencing, and retransmission.
- On Data Link layer
detects errors using
CRC and controls frame
delivery.

Step 1: Number of segments.

$$= \text{file size} \div \text{segment size}$$

$$= 157,286,400 \div 1500$$

$$= \boxed{104,858} \text{ segments}$$

Step 2: Number of frames

$$= \text{Number of segments} = \boxed{104,858} \text{ frames}$$

Step 3: Total overhead:

$$= 104,858 \times 40$$

$$= \boxed{4,194,320} \text{ bytes}$$

Step 4: Total data sent

$$= 157,286,400 + 4,194,320$$

$$= \boxed{161,480,720} \text{ bytes}$$

Step 5: Transmission Time.

$$= (161,480,720 \times 8) \div 50,000,000$$

$$= \boxed{25.845}$$

② sliding window

Given

$$\rightarrow \text{window size} = 6$$

$$\rightarrow \text{frames} = 48$$

Step 1: windows Required:

$$= 48 \div 6$$

$$= 8 \text{ windows}$$

STEP 2: Retransmissions

$$= 1 \text{ lost frame} \times 8 \text{ windows} \\ = 8 \text{ retransmissions}$$

$$\therefore \boxed{\text{windows} = 8} \\ \boxed{\text{Retransmissions} = 8}$$

Flow control: It prevents sender from overloading the receiver, impact efficiency

③ IP Fragmentation.

Given:

-7 Packet = 121000 bytes

-7 MTU = 1500 bytes

-7 Header = 20 bytes.

STEP 1: Data per fragments

$$= 1500 - 20$$

$$= \boxed{1480 \text{ bytes}}$$

STEP 2: Number of fragments

$$= 121000 \div 1480$$

$$= \boxed{81 \text{ fragments}}$$

STEP 3: Header overhead

$$= 81 \times 20$$

$$= \boxed{1620 \text{ bytes}}$$

④ Sliding window:

STEP 1: windows

$$= 48 \div 6 = \boxed{8}$$

STEP 2: Retransmissions

$$= \boxed{8}$$

Network layer: Uses fragment offset and identification fields so the destination can correctly reassemble fragments.

⑤ session layer:-

Given:-

$$\rightarrow \text{File} = 600 \text{ MB}$$

$$\rightarrow \text{check point} = 60 \text{ MB}$$

$$\rightarrow \text{Failure} = 390 \text{ MB}$$

Step 1: check points

$$= 390 \div 60$$

$$= 6 \text{ checkpoints}$$

Step 2: Last check point

$$= 60 \times 6$$

$$= 360 \text{ MB}$$

Step 3: Retransmitted Data

$$= 390 - 360$$

$$= 30 \text{ MB}$$

$\therefore$ check points = 6.

Retransmitted Data = 30 MB

⑥ Presentation layer:

Given

$$\rightarrow \text{original} = 900 \text{ MB}$$

$$\rightarrow \text{compression} = 40\%$$

$$\rightarrow \text{Encryption} = 5\%$$

Step: 1: compressed size

$$= 900 \times 60\%$$

$$= 540 \text{ MB}$$

Importance:-

synchronization
allows communication to
resume from the last
checkpoint instead of
restarting.

Step 2: Encrypted size

$$= 540 \times 1.05$$

$$= 567 \text{ MB}$$

Step 3: Reduction

$$= (1900 - 567) \div 900 \times 100$$

$$= 37\%$$

$\therefore$ Compressed size = 540 MB

Encrypted size = 567 MB

Reduction = 37%

⑦ FTP Transfer:

Given:

- Data = 3 GB = 3072 MB

- Request = 3 MB

- Speed = 120 Mbps

Step 1: Requests

$$= 3072 \div 3$$

$$= 1024$$

Step 2: Transmission Time.

$$= (3072 \times 8) \div 120$$

$$= 204.8 \text{ s}$$

$\therefore$

Requests = 1024

Time = 204.8 s.

Application Layer provides file transfer, user authentication, and error reporting.

End to End Delay
Given:

$$\text{processing} = 4 + 3 + 5 + 2 + 1 \text{ ms}$$

$$\text{propagation} = 18 \text{ ms}$$

$$\text{transmission} = 12 \text{ ms}$$

Step 1: processing delay

$$= 4 + 3 + 5 + 2 + 1$$

$$= 15 \text{ ms}$$

Step 2: Total Delay

$$= 15 + 18 + 12$$

$$= 45 \text{ ms}$$

$$\therefore \text{Total delay} = 45$$

Layer roles:

- Application - user services
- Transport - reliability
- Network - routing
- Data Link - framing
- Physical - transmission.

Q Frame Efficiency

Given:

$$\rightarrow \text{Useful Data} = 80 \text{ MB}$$

$$\rightarrow \text{Frame} = 1600 \text{ bytes.}$$

$$\rightarrow \text{Overhead} = 80 \text{ bytes}$$

Step 1: Useful Data per frames.

$$= 83,886,080 \div 1520$$

$$= 55,188$$

Step 2: Number of frames

$$= 83,886,080 \div 1520$$

STEP 3: overhead

$$= 55,188 \times 80 = \boxed{4,415,040 \text{ bytes}}$$

STEP 4: Efficiency

$$= (1520 \div 1600) \times 100$$

$$= \boxed{95\%}$$

Effect:

Higher protocol overhead reduces bandwidth available for actual data

⑩ Hospital medical Record:-

Given

-> File = 240 MB

-> segment = 1400 bytes

-> compression = 30% -> overhead = 60 bytes

STEP 1:- compressed size

$$= 240 \times 70\%$$

$$= \boxed{168 \text{ MB}}$$

STEP 2:- convert to Bytes

$$= 168 \times 1024 \times 1024$$

$$= \boxed{176,160,768 \text{ bytes}}$$

STEP 3:- segments

$$= 176,160,768 \div 1400$$

$$= \boxed{125,829}$$

STEP 4:- Protocol overhead

$$= 125,829 \times 60$$

$$= \boxed{7,549,740 \text{ bytes}}$$

STEP 5:- final Data

$$= 176,160,768 + 7,549,740$$

$$= \boxed{183,710,508 \text{ bytes}}$$

OSI cooperation:- The presentation layer compresses and encrypts data, the transport layer segments it and the data link layer detects error.